

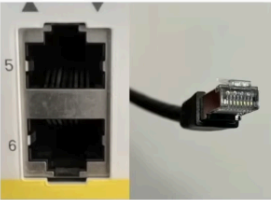
Day 3: How the TCP/IP Model Actually Works

TCP/IP Model

Layer 1: The Physical layer

Layer 1: The Physical layer

- The **Physical Layer** (Layer 1) sends and receives bits as electrical, optical, or radio signals over the medium.
- Defines things like cables, connectors, signal levels, and link speeds.
- Examples: copper UTP cables, fiber-optic cables, Wi-Fi radios and antennas, network interface cards (NICs).
- The physical aspects of transmitting data are very complex.
 - Network engineers typically don't have to know the low-level details.



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Application layer

Transport layer

Internet layer

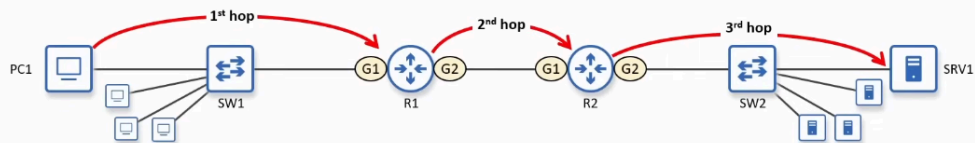
Local Network layer

Physical layer

Layer 2: The Data Link layer or Local Network layer



Layer 2: The Local Network layer



- The **Local Network layer** (Layer 2) provides **hop-to-hop** delivery of messages on a local network.
 - A **hop** is one step along the path between two devices:
 - from one router or host, to the next router or host in the path.
 - Switches don't count: a switch just extends the local network, allowing multiple devices to connect.
- Uses **MAC (Media Access Control) addresses** to identify interfaces.
 - PC1** sends the message to the MAC address of **R1's G1 interface (NIC)**.
 - R1** sends the message to the MAC address of **R2's G1 interface (NIC)**.
 - R2** sends the message to the MAC address of **SRV1's interface (NIC)**.
- Protocols at this layer include:
 - Ethernet (IEEE 802.3)
 - Wi-Fi (IEEE 802.11)

Application layer

Transport layer

Internet layer

Local Network layer

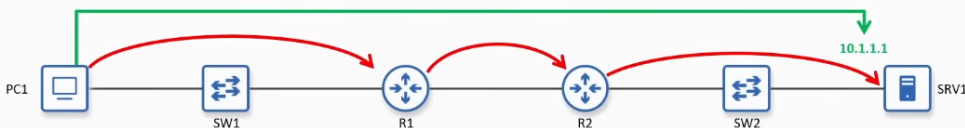
Physical layer



Layer 3: The Network layer or Internet layer



Layer 3: The Internet layer



- The **Internet layer** (Layer 3) provides end-to-end delivery between hosts across multiple networks.
 - Internet = internetwork (between networks)
- Uses **IP addresses** to identify hosts in the network.
 - When PC1 sends a message to SRV1, it addresses the message to SRV1's IP address.
- Routers** operate mainly at this layer, using the message's destination IP address to forward the message toward its final destination host.
- Protocols at this layer include:
 - IP (IPv4, IPv6)
 - ICMP (Internet Control Message Protocol)

Application layer

Transport layer

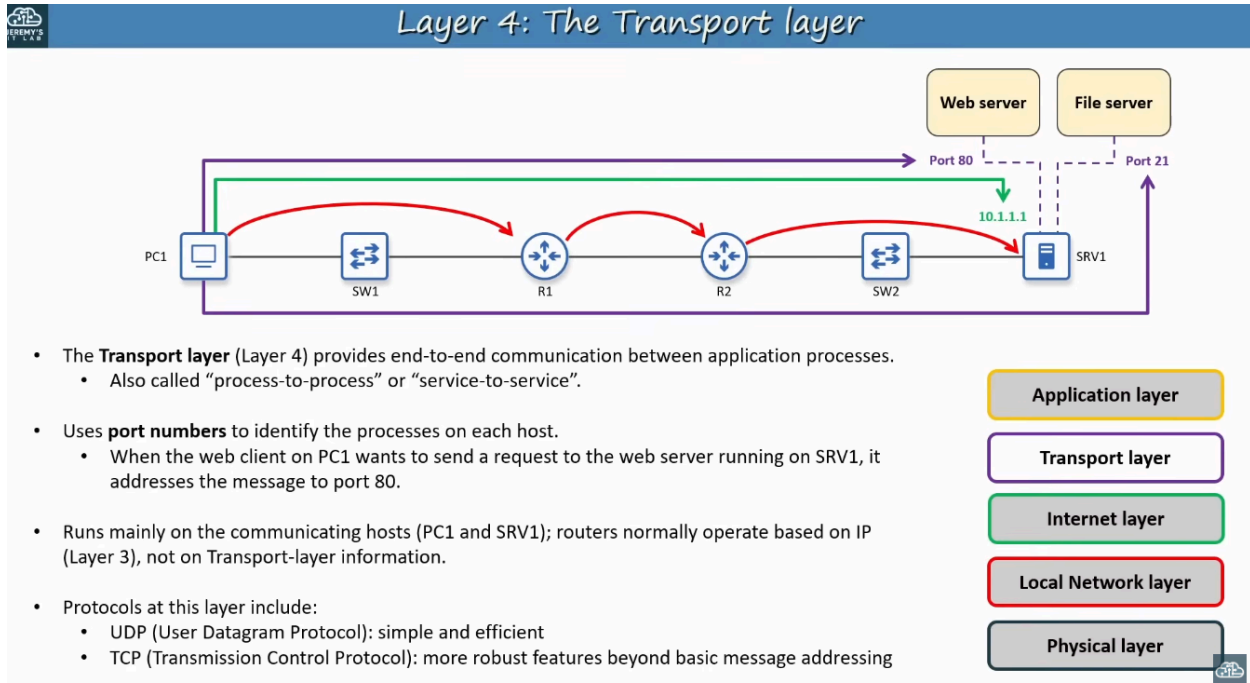
Internet layer

Local Network layer

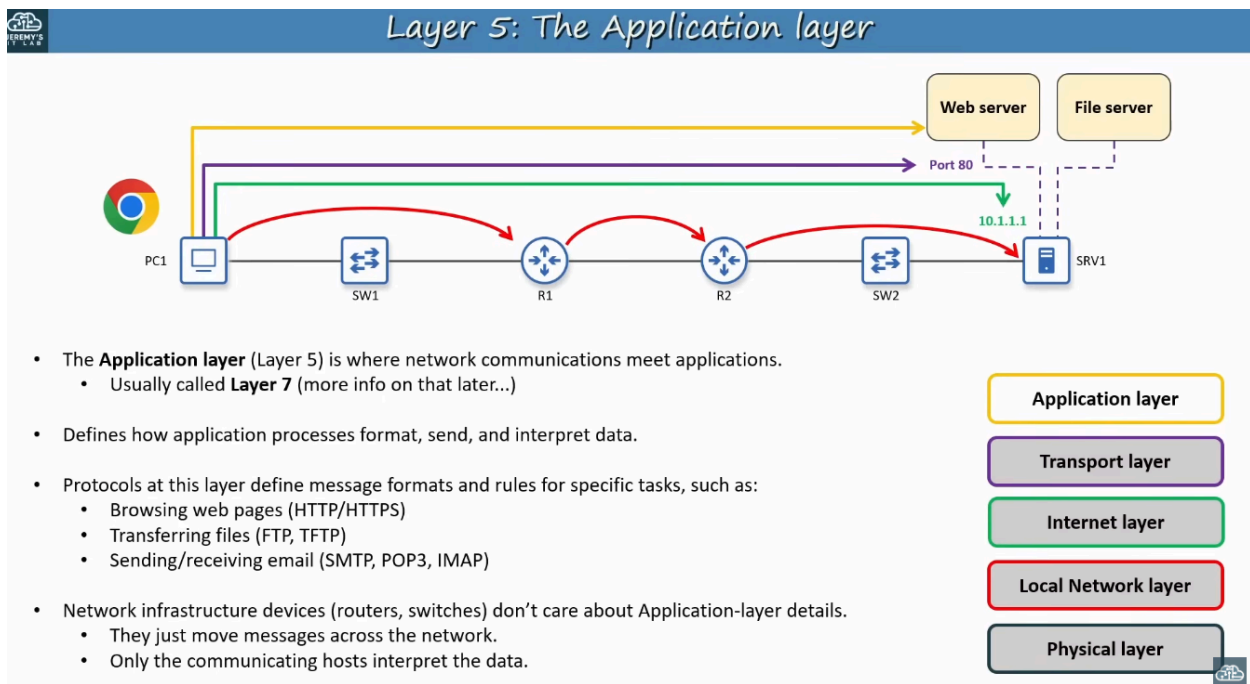
Physical layer



Layer 4: The Transport layer



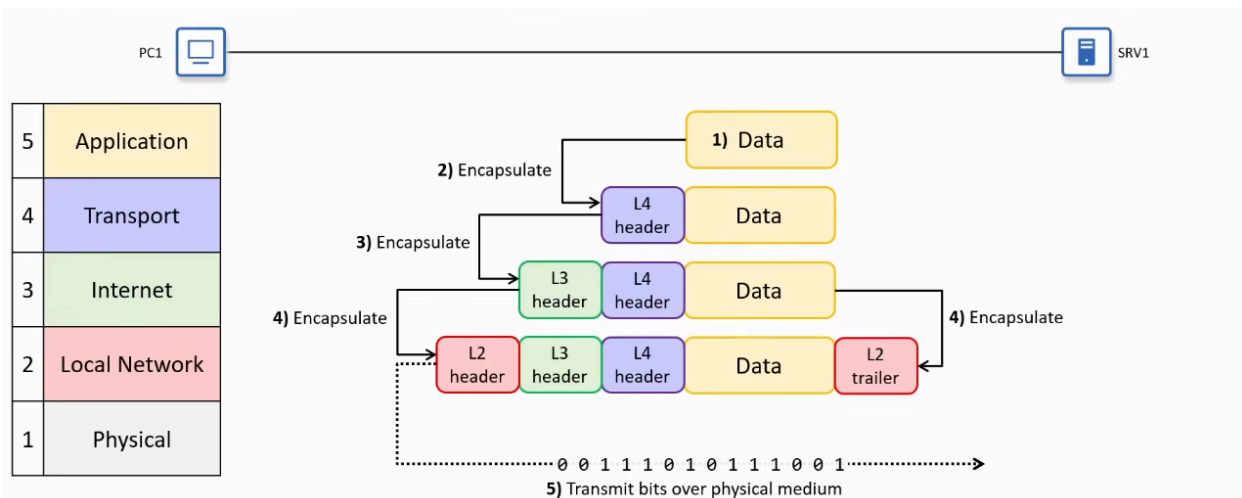
Layer 5: The Application layer



Encapsulation and Decapsulation

Encapsulation Process

- The **Application layer** prepares the data to be sent over the network.
- As the message moves down the stack, each layer **ENCAPSULATES** the data with a **HEADER** including the information needed for that layer.
 - Source and destination addresses (port numbers, IP addresses, MAC addresses), etc.
 - Layer 2 also adds a **TRAILER** that the receiving device uses to check for transmission errors.
- The **Physical layer** transmits the bits as signals over the physical medium.
 - The L2 **HEADER** is transmitted first, and the L2 **trailer** is transmitted last.

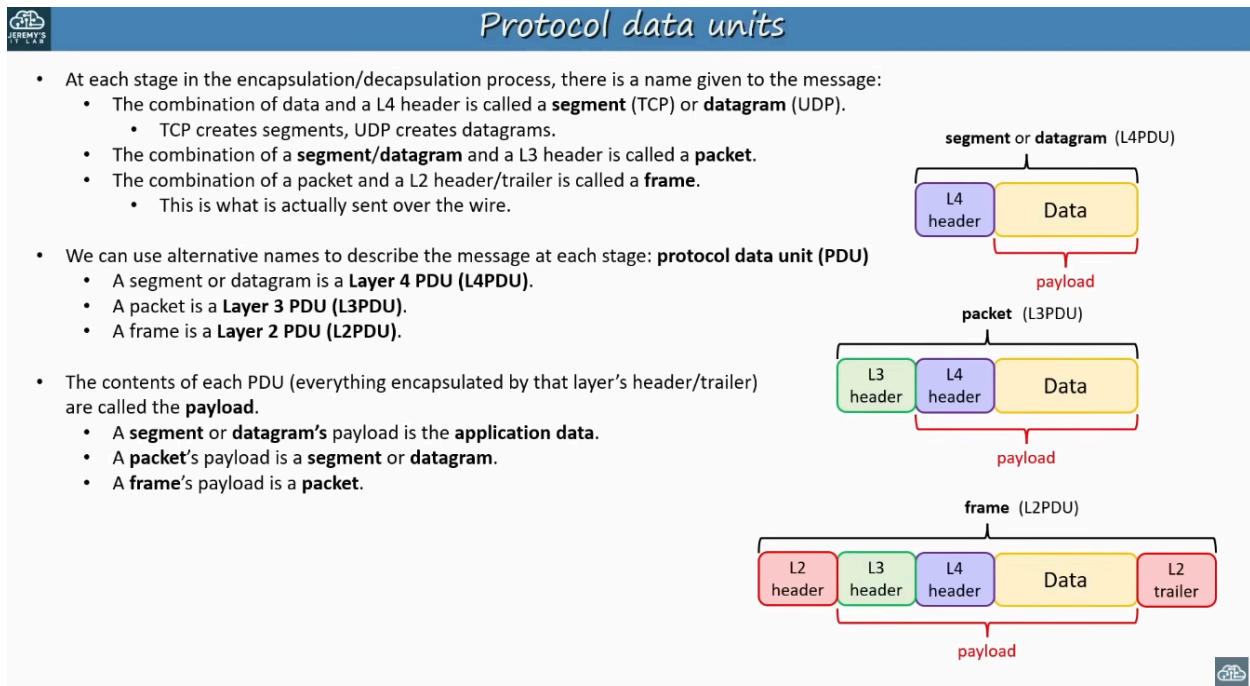


Decapsulation Process

- The receiving device receives the message as a stream of bits at layer 1.
- The device examines the information in the Layer 2 header and trailer, and then removes them (decapsulation).

- The decapsulation process continues up the stack: Layer 3 removes the L3 header, then Layer 4 removes the L4 header, and then the data is delivered to the Application layer.
- The application processes the data and, if needed, generates a response that goes back down the stack.

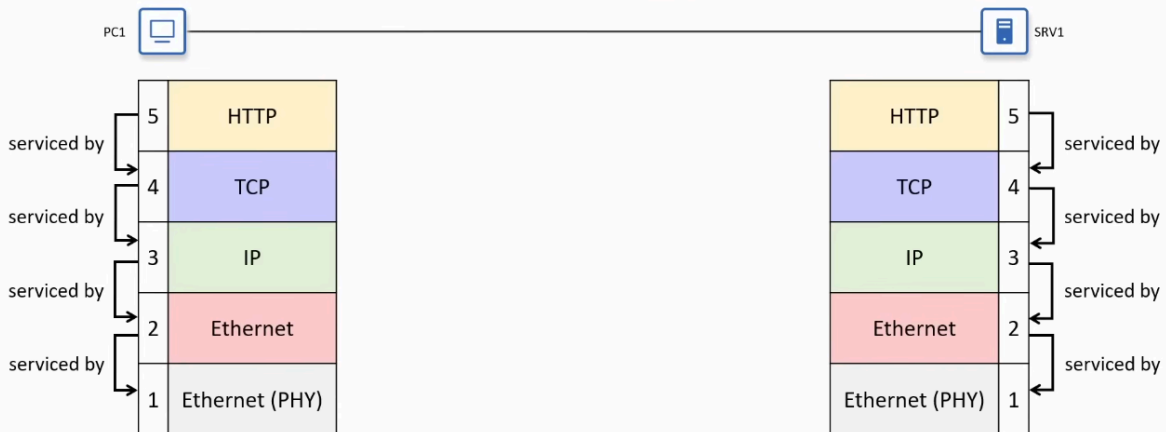
Protocol data units



Note: The payload is what inside the PDU, not including Layer Header or Trailer.

Adjacent-layer interaction

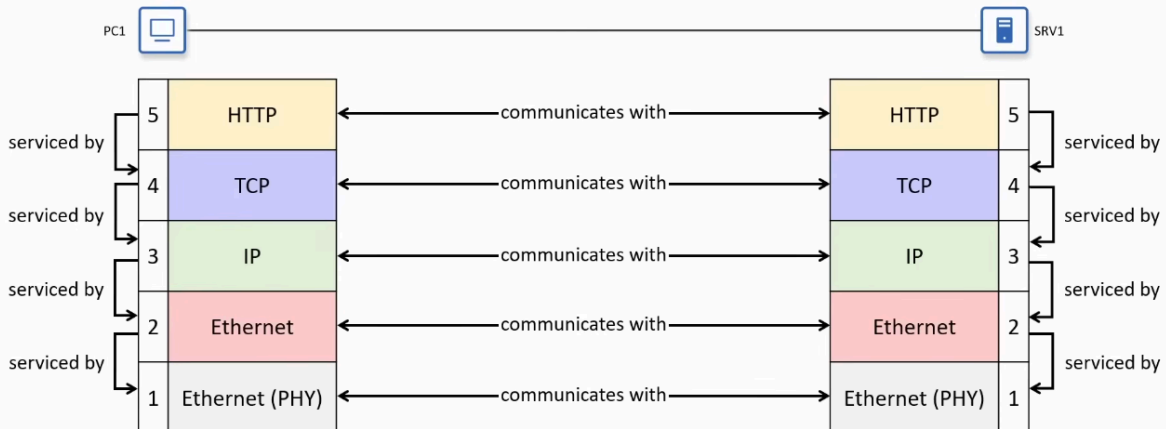
Adjacent-layer interaction



- Each layer provides a service to the layer above it, and is serviced by the layer below it (**adjacent-layer interaction**).
 - Layer 4** provides a service to **Layer 5** by delivering data to the correct application using port numbers.
 - Layer 3** provides a service to **Layer 4** by delivering segments/datagrams to the correct destination host using IP addresses.
 - Layer 2** provides a service to **Layer 3** by delivering packets to the next hop using MAC addresses.
 - Layer 1** provides a service to **Layer 2** by sending and receiving frames as electrical, optical, or radio signals.

Same-layer interaction

Same-layer interaction

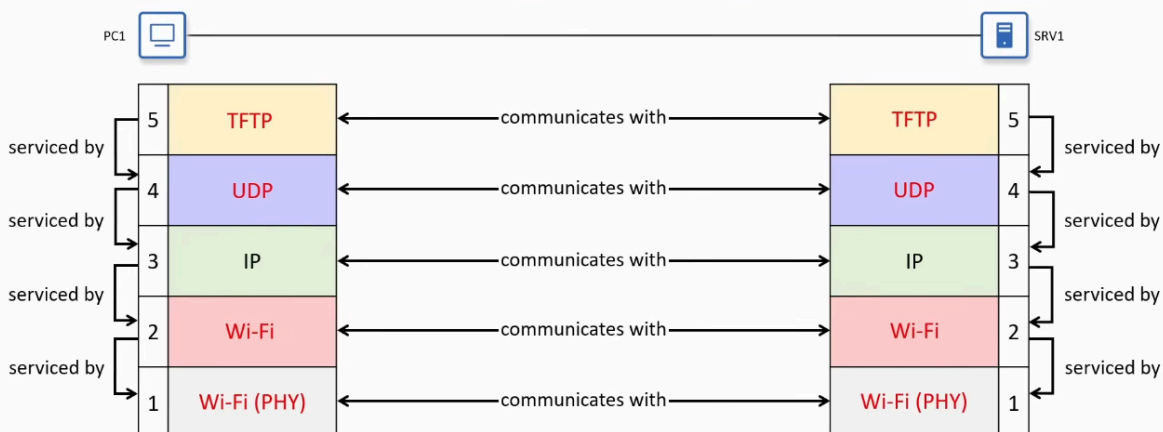


- Each layer communicates with the same layer on other devices (**same-layer interaction**).
 - The Application layer on one host sends data to the Application layer on the other host.
 - A segment/datagram is addressed to the **Layer 4 port number** of the correct application on the destination host.
 - A packet is addressed to the **Layer 3 IP address** of the destination host.
 - A frame is addressed to the **Layer 2 MAC address** of the next hop.
 - Signals sent out of a physical port are received by a **physical port** on the connected device.



Separation of layers

Separation of layers



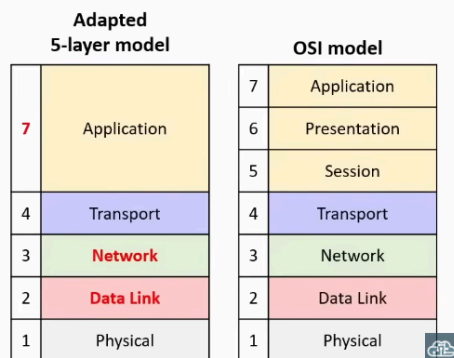
- Each layer has its own job and provides a specific service to the layers above.
 - The layers are modular.
- As long as each layer keeps its “contract” with the other layers, we can improve or replace protocols at different layers without redesigning everything.
- That flexibility is one of the main benefits of a layered model.



The OSI model

The OSI model

- TCP/IP development started in the 1970s (ARPANET work, early TCP/IP specs).
- In the late 1970s and 1980s, the **International Organization for Standardization (ISO)** designed a 7-layer **Open Systems Interconnection (OSI) model** and a matching protocol suite.
 - The goal was to create international, vendor-neutral networking standards that could unify existing proprietary stacks and potentially replace TCP/IP.
- Governments, including the US, promoted OSI as the preferred/recommended stack for new deployments.
- OSI protocols ended up being late and complex, and never gained the same deployment as TCP/IP.
 - TCP/IP “won” in the real world, although some OSI technologies are still used.
- Today, almost all real networks use TCP/IP, but the 7-layer OSI model survives as a reference/teaching model and a common way to talk about “layers”.
- Most networking resources use a 5-layer model like the one covered in this video, but with names from the OSI model.
 - That is why the TCP/IP Application layer is often called **Layer 7**.



Other versions

Arpanet Reference Model (RFC 871)	Internet Standard (RFC 1122)	Internet model (Cisco Academy ^[56])	TCP/IP 5-layer reference model (Kozierok, ^[57] Comer ^[58])	TCP/IP 5-layer reference model (Tanenbaum ^[59])	TCP/IP protocol suite or Five-layer Internet model (Forouzan, ^[60] Kurose ^[61])	TCP/IP model (Stallings ^[62])	OSI model (ISO/IEC 7498-1:1994 ^[63])
Three layers	Four layers	Four layers	Four+one layers	Five layers	Five layers	Five layers	Seven layers
Application/ Process	Application	Application	Application	Application	Application	Application	Application
							Presentation
							Session
Host-to-host	Transport	Transport	Transport	Transport	Transport	Host-to-host or transport	Transport
	Internet	Internetwork	Internet	Internet	Network	Internet	Network
Network interface	Link	Network interface	Data link (Network interface)	Data link	Data link	Network access	Data link
—	—	—	(Hardware)	Physical	Physical	Physical	Physical

